

How much of a nervous-system model does behaviour identify?

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Abstract

Whole-organism simulators reproduce behaviour from thousands of biophysical parameters; which of them behaviour determines is unknown. Perturbing each of a nematode model’s 3076 connectome weights, read through more observables than parameters, behaviour determines eight directions at 90% of the curvature and 47 at 99%. A *Drosophila* optic-lobe model agrees at 14 of 330: in two phyla, behaviour reaches a few per cent of a connectome-scale space. Chemical synapses carry 87% of the curvature, gap junctions 13%, and the free conductances, fast voltage-gated calcium and calcium-activated potassium, are those biology permits to vary. Cell models with known ground truth verify the geometry: small neuronal models are recovered exactly, and the MAPK cascade matches behaviour with its parameters wrong. Further aspects of one action add little: their subspaces overlap far more than they complement. What behaviour leaves free is the model’s remaining degrees of freedom, and we locate them.

Introduction

A central premise of mechanistic whole-organism simulators is that fitting them to behaviour recovers the biophysics that produces it. Decades of work on conductance-based models, however, from the stomatogastric ganglion (1, 2) to the broader theory of sloppy models (3, 4), establish that many parameter sets produce indistinguishable dynamics: the map from parameters to behaviour is many-to-one (5–7). Whether, and how strongly, this degeneracy holds at the scale of a *whole-organism* nervous system, with thousands of synaptic, junctional and channel parameters driving a behaving body, has not been measured. The consequences are direct: if behaviour constrains only a low-dimensional subspace, then behaviour-based calibration can recover that subspace but not the remainder, and the parameters it leaves unconstrained are candidates for the directions along which biology can vary without behavioural consequence.

Here we measure the behavioural-equivalence manifold of four published whole-organism motor simulators (BAAIWorm (8), modWorm (9), flybody (10), and NeuroMechFly v2 / FlyGym (11, 12)), which are built on physics engines for musculoskeletal control (13). To test generality, we apply the same measurement to a connectome-constrained *Drosophila* visual circuit, the canonical crustacean pyloric population, and fly-larva and mammalian controllers; and to verify it against ground truth, to eleven cell models whose parameters are known exactly, spanning neuronal excitability and intracellular signalling. We measure the manifold of behaviourally equivalent parameter sets, identify the constrained (stiff) subspace, and ask whether its low-dimensional geometry recurs across systems and in biology. The contribution is a measurement verified where truth is knowable, together with convergent observations from biological data that use none of these models.

The components required to do this at scale are now available. Connectome-constrained whole-organism simulators rest on increasingly complete wiring diagrams (14–16) on community simulation frameworks (17–19), and on the whole-brain neural recordings they aim to reproduce (20–22); high-throughput tracking reduces behaviour to low-dimensional kinematic descriptors (23–25); and parameter inference is increasingly amortised by simulation-based methods (26–28) or performed by black-box optimisers (29, 30); behaviour-based parameter generation for whole-organism simulators has recently been demonstrated from tracking video alone (31), and causal-model estimation for the fly brain argues that its dominant dynamics are carried by a small fraction of the wiring (32). What none of these components settles is *how much* of a whole-organism parameter set behaviour actually determines: the classical identifiability question (33, 34) that sloppy-model theory answers for small systems (3, 4, 35) and that pyloric degeneracy answers for a single circuit (1, 2). Quantifying it at whole-organism scale is a prerequisite for behaviourally grounded digital organisms (36, 37).

Results

To measure what behaviour determines in a whole-organism model we need a system in which the forward map from biophysics to behaviour is fully specified and cheap enough to probe densely. We therefore work with published whole-organism simulators, in which every synaptic, junctional and channel parameter is explicit and behaviour can be re-simulated under controlled perturbation. Our primary system is the nematode, for which two independently constructed models exist on the same connectome, so that any geometry we measure can be checked against a second implementation of the same animal. Around this core we assemble a comparison set spanning four orders of parameter count, and apply one estimator to all of them (Fig. 1).

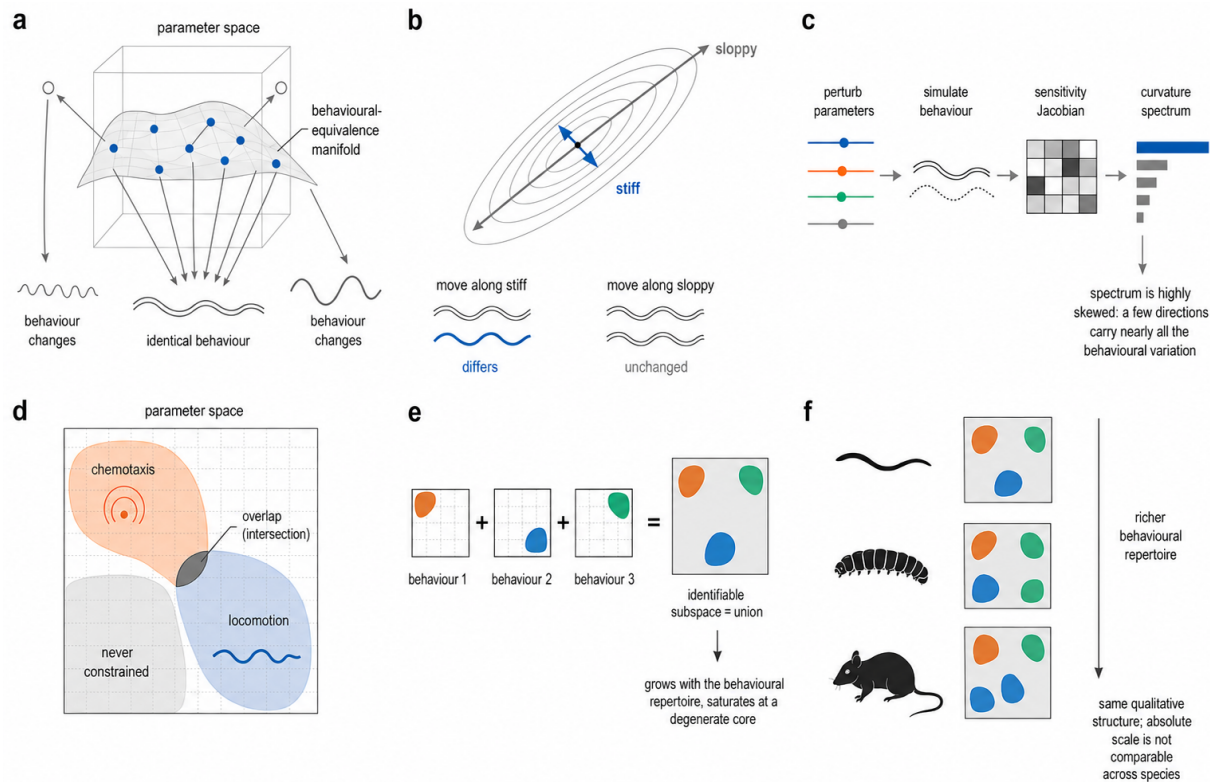


Figure 1: What behaviour can and cannot determine about a nervous-system model.

a Many distinct parameter sets produce the same behaviour. Blue points lying on a curved sheet within parameter space (the behavioural-equivalence manifold) all generate one identical behavioural trajectory; points off the sheet (open circles) generate visibly different behaviour. **b** The manifold is anisotropic. Contours of the behavioural loss are elongated: motion along a short, high-curvature axis (blue, stiff) changes behaviour, whereas motion along a long, low-curvature axis (grey, sloppy) leaves it unchanged. **c** The measurement. Each parameter is perturbed, behaviour is re-simulated, the sensitivity Jacobian is assembled and its Gauss–Newton curvature spectrum is diagonalised; the spectrum is strongly skewed, so a small number of directions carries nearly all behavioural variation. **d** The geometry is checked against truth where truth is known. In cell models with a known parameter vector (star), a fitted parameter set far from the truth (blue point) can produce behaviour indistinguishable from it (upper trace), while a displacement of the same size off the equivalence manifold changes the behaviour outright (vermillion, lower trace): a perfect behavioural fit does not imply recovered parameters, and where behaviour does identify them, recovery is exact. **e** What the paper measures within one organism. Two aspects of a single action (blue, vermilion) each determine a region of parameter space, and those regions overlap across most of their area, so measuring a further aspect largely re-measures the same combinations. Whether genuinely different actions behave the same way (dashed outline) is not testable in the simulators available and is left empty here. **f** What is left determined and what is left free is biophysically specific: the wiring is determined (blue), the fast voltage-gated conductances are free (grey). The panels state the quantities measured, not their magnitudes; the sheet in **a** and the regions in **e** are schematic. Colour is used consistently throughout the paper: blue, stiff or identifiable; grey, sloppy or unconstrained; vermillion and green, distinct behaviours.

Behaviour converges while parameters remain dispersed

We first asked how much of a connectome-scale model behaviour can reach. On the BAAIWorm nematode model (136 cells; the 3076 synaptic and gap-junction weights of the connectome) we perturbed every weight in turn and read the motor output as 9600 observables, so that observables outnumber parameters and the curvature rank is not capped by the measurement. The behavioural Gauss–Newton curvature has effective dimension 8 at 90% and 47 at 99% of 3076, with participation ratio 2.55: of three thousand connection directions, behaviour constrains eight. Only 4 weights produce no measurable change, so the flat directions are degeneracy rather than unconnected parameters (Fig. 2a). The measurement is differentiable and noise-free here: the response is proportional to the step to four significant figures over a twentyfold range, and repeat simulations are bit-identical (Supplementary §1.5). Nor is it tied to one parameter point: at four displaced operating points the effective dimension stays at 8–12 and the stiff subspaces coincide (cosine 0.89, random 0.15; Supplementary §5.3).

The behavioural-equivalence manifold is low-dimensional

The same geometry appears at every parameter scale, provided the observable count does not limit the answer. A Gauss–Newton Hessian from n_{obs} observables has rank at most n_{obs} , so a low effective dimension read through a few summary statistics says as much about the measurement as about the model. On the 48 FlyGym actuator gains, ten summary scalars give 1, whereas 4200 joint-angle observables give 16 (90%) / 38 (99%) against a noise floor of 7 (Supplementary §5.1); on modWorm the answer does not move between six scalars and 7,200 posture observables. Across parameter counts of 4, 12, 48 and 3076, behaviour is set by a low-dimensional stiff subspace embedded in a high-dimensional flat manifold (Fig. 4b).

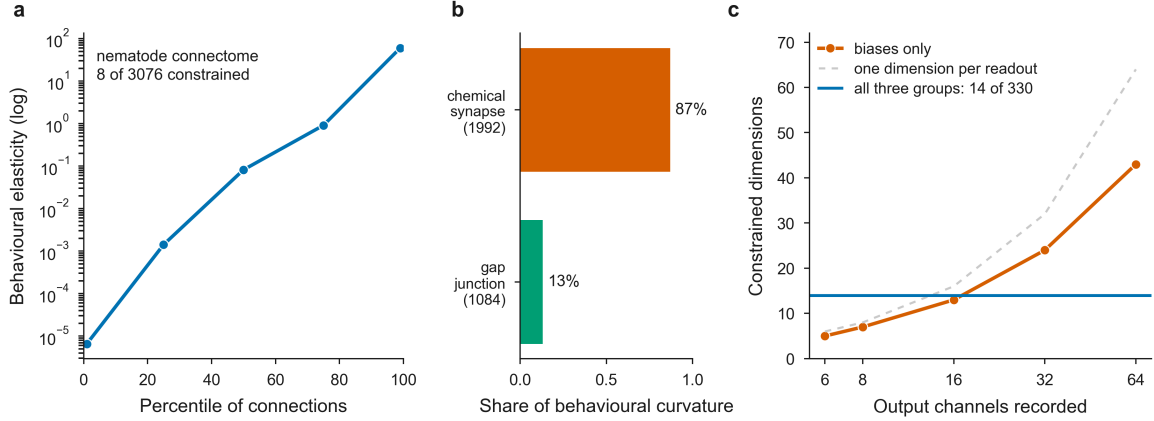


Figure 2: **What behaviour constrains in two connectome-scale models, and what sets the number.** **a** Behavioural sensitivity of the 3076 synaptic and gap-junction weights of the BAAIWorm nematode connectome, perturbed individually and read through 9600 motor-output observables so that observables outnumber parameters. Points are the stored percentiles of the per-connection elasticity, which span seven orders of magnitude. Effective dimension 8 (90%) / 47 (99%), participation ratio 2.55: 3068 directions are flat, and only 4 weights have no measurable effect at all. **b** The same curvature split by connection type. Chemical synapses carry 87% of it from 1992 connections and gap junctions 13% from 1084, so per connection a chemical synapse contributes about five times more. **c** The flyvis *Drosophila* optic-lobe model. Vermillion: effective dimension of the 65 per-cell-type biases as the number of recorded output channels is reduced from 64 to 6; grey dashed: one constrained dimension per readout. Blue line: the model measured over all three of its free parameter groups, 14 of 330 sampled parameters. A nematode motor circuit and a fly visual circuit thus agree that behaviour reaches a few per cent of a connectome-scale parameter space, and the bias-only figure of 43 of 65 tracks the readout count rather than the circuit: an effective dimension measured through fewer observables than parameters is a statement about the measurement. Both models pass a differentiability check (the response is proportional to the perturbation over a twentyfold range in the nematode and across two decades in the fly), and repeat nematode simulations are bit-identical, so the finite-difference noise floor there is zero. Curvature is the Gauss–Newton form $H = J^T W J$; effective dimension is the smallest number of eigenvalues reaching the stated fraction of the spectral mass.

The manifold is the neural control of eigenworm posture

The nematode behavioural manifold is not an abstract low-dimensional set but the parameter-space image of the animal’s known approximately four-dimensional postural repertoire. Real *C. elegans* posture from the OpenWorm Movement Database occupied an effective 3.79–4.14 dimensions, with four eigenworms capturing 94–96% of postural variance, reproducing the classical eigenworm decomposition (23). Two independently constructed worm models generate posture that overlaps this empirical basis, at mean principal-angle $\cos 0.77$ for modWorm and $\cos 0.68$ – 0.69 for BAAIWorm. Two qualifications bound that overlap: the models’ postural repertoires are lower-dimensional than the animal’s (1.66 and 2.76–3.30 effective dimensions against 3.79–4.14), and the agreement is a property of the subspace rather than of individual modes, whose alignments are 0.30, 0.02, 0.57 and 0.24. These alignments substantially exceeded those of random four-dimensional subspaces (modWorm $p=0.001$, BAAIWorm $p=0.04$ even within a conservative ten-dimensional posture ambient; $p<10^{-4}$ in the full space; Supplementary §6.1), and within each model it was the stiff mechanisms,

not the sloppy ones, that displaced the eigenworm trajectory. In the systems examined here the effective rank of behavioural sensitivity tracks the intrinsic dimensionality of the postural repertoire, rather than the number of observables recorded: re-measuring with 7,200–14,400 posture observables in place of six summary kinematics left the effective dimension at 2–3 of 7 (Supplementary §6.2). We note that this is an empirical property of these models and behavioural losses, not a general bound: a low-dimensional attractor can in principle constrain many parameter combinations through transients, multiple initial conditions or nonlinear temporal structure, and the intrinsic dimensionality of the behaviour need not equal the rank of the parameter-to-trajectory Jacobian. The identifiable subspace is therefore related to, but not identical with, the neural control of the animal’s postural repertoire (Fig. 3a).

Aspects of one action are largely redundant

A low-dimensional behaviour raises the question of how much a repertoire could add. Two things are usually conflated: aspects of one action, and different actions. Only the first is testable in available simulators. In BAAIWorm, perturbing the chemosensory parameters altered behaviour $7.15\times$ more during chemotaxis (food-directed; sensitivity 0.872) than during plain locomotion in a weak gradient (0.122), whereas perturbing the motor parameters showed the opposite contrast (0.109 versus 0.357; Supplementary §6.3): the axis a behaviour renders stiff is the axis its function requires, and chemotaxis and locomotion render *different* axes stiff. This matches the anatomical separation of the chemosensory navigation circuit from the ventral-cord locomotor circuit (38, 39), and models in which distinct premotor and motor elements generate the rhythm (40–42). That contrast is between two functions of the same circuit. Between aspects of one action the picture inverts: the subspaces they determine overlap far more than they complement, so further aspects largely re-measure the same combinations. That redundancy is degeneracy, measured without a threshold. We measured the residual gain against explicit nulls. Four distinct locomotor presets each constrained 2–4 of 7 mechanism directions at mean pairwise stiff-subspace $\cos 0.54$, where randomly oriented directions in the same space give $\cos 0.31$ and span 3.50 dimensions (95% interval 3.00–3.88), and a single shared direction would give $\cos 1$ and span 1. The observed subspaces are significantly more aligned than chance ($p=0.016$ for the cosine, $p=0.020$ for the span, 20,000 draws) while still spanning 2.98 of 4: aspects of an action are markedly more redundant than random directions, without collapsing onto a single one (Supplementary §6.4). Their union reached 5 of 7 at 99% of the curvature and 2 at 90%; across twelve behaviours it saturated at 4 and 2, leaving a residual 2–3 directions that no behaviour constrains, an irreducible degenerate core. The stiff and sloppy directions are distributed parameter combinations rather than single parameters (stiff-subspace participation 3.42, sloppy 4.68 of 7; Supplementary §6.7), so this core is a genuine biological neutral space and not an unused parameter (Fig. 3b,c).

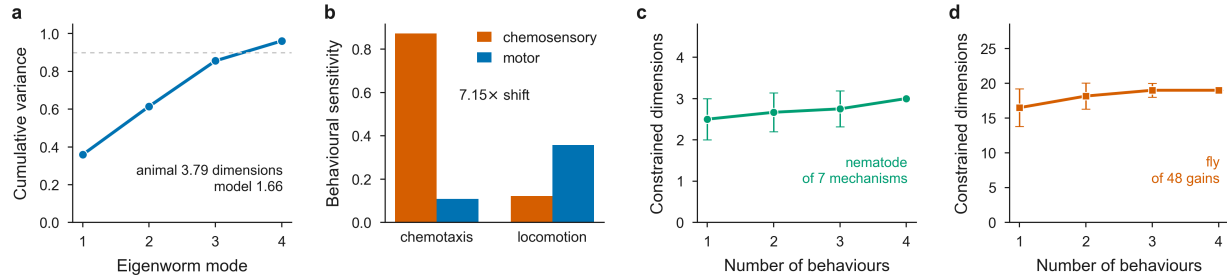


Figure 3: The postural basis, the context dependence of sensitivity, and what a larger behavioural repertoire adds. **a** Cumulative postural variance of real *C. elegans* across eigenworm modes (OpenWorm Movement Database, $n=8432$ frames); four modes capture 0.96 of the variance and the animal’s posture occupies 3.79 effective dimensions, against 1.66 for the model Worm. **b** The same parameters are stiff in one behaviour and sloppy in another. Vermillion: chemosensory parameters; blue: motor parameters. Chemosensory sensitivity is 0.872 during food-directed chemotaxis and 0.122 during plain locomotion, a $7.15\times$ shift, while the motor parameters show the reverse contrast (0.109 against 0.357). **c**, **d** Constrained dimensions against the number of behaviours, in the nematode over its mechanism classes and in the fly over its forty-eight actuator gains, each read through trajectory-level observables. Points average over every subset of that size, so the curve does not depend on one ordering; error bars are the spread across subsets. Adding behaviours does enlarge the identifiable subspace (2.50 to 3.00 in the nematode, 16.5 to 19.0 in the fly), but the gain saturates by the third or fourth behaviour and the spread falls to zero, so behaviours overlap far more than they complement: four fly behaviours give 19 rather than four times the 16.5 that one gives. All behaviours compared here share a triggering mode. The two panels keep separate axes because the models are parameterised differently and their absolute dimensions are not comparable. Effective dimension is the smallest number of eigenvalues reaching 90% of the spectral mass of the positive semi-definite Gauss–Newton curvature.

The same geometry in a second phylum

The organisation is not specific to the nematode. Applying the same probe to the FlyGym walking controller gives a stiff subspace of 16 of 48 actuator gains read through 4200 joint-angle observables, and adding behaviours moves it to 19 (Supplementary §6.5). A fly larva and a mammalian controller, probed with the same estimator, show the same qualitative pattern; their observable sets are smaller than their parameter counts, so we read structure from them and not absolute dimensions. Absolute dimensions are in any case not comparable across these models, which are parameterised differently; the structure and not the count is what recurs.

Ground truth verifies the geometry where it can be known

The whole-organism models have no true parameter vector; at the cell scale the curvature can be checked against truth. Across eleven cell models measured identically and recovered from random starts independent of the truth, both ends of the geometry appear: the 2–4-parameter neuronal models are recovered exactly (error $< 10^{-4}$), while in the canonical MAPK cascade (43), the family in which sloppiness was first described (3), behaviour is matched to 10^{-3} while the parameters end 40% from the truth, farther than they started: a perfect behavioural fit does not imply recovered parameters. Truth also bounds the effective dimension’s meaning: across a parameter-count $\times$ observable grid on one neuron it is 1 in every cell while achieved recovery error spans

4–56%; it measures curvature concentration, not an upper bound on recoverable parameters (Supplementary §7). On the worm model an independent multi-start probe ranks the identifiable axes identically ($\rho=0.96$, $p=5 \times 10^{-4}$), with simulation-based calibration as a control.

Stiff and sloppy directions are biophysically interpretable

Decomposing the BAAIWorm behavioural curvature over the model’s named conductances, read through the full motor-neuron trajectory rather than a handful of scalars, gives an effective dimension of 3 (90%) / 6 (99%) over the twelve mechanisms that have a measurable effect, with elasticities spanning three orders of magnitude. The partition by mechanism is clear: the stiffest axes are the inward rectifier IRK, the sodium leak NCA and the potassium conductances EGL-36 and EGL-2, whereas the fast voltage-gated calcium channels EGL-19 and UNC-2 are among the sloppiest and the calcium-activated potassium channels SLO-1 and SLO-2 produce no measurable change in behaviour at all (Supplementary §5.2). Behaviour is thus determined by how cells are connected and by their passive and slow properties, and is largely insensitive to the fast voltage-gated calcium conductances, consistent with the known compensation and variability of these currents in small circuits (2). Most of the degeneracy lies in the connectome: 3068 of 3076 directions are flat, and the remaining curvature is uneven across connection types: 1992 chemical synapses carry 87% of it and 1084 gap junctions 13%, so per connection a synapse matters about five times more (Supplementary §5.3). Per-connection elasticities span seven orders. This partition is not an artefact of one model: probed identically, two independently constructed worm simulators—modWorm (coupled-ODE, 7 mechanisms) and BAAIWorm (NEURON, 20 named channels)—agree that the network-connectivity axes are stiff (stiff fraction 0.83 / 0.87) while the single-cell biophysical axes are sloppy (0.08 / 0.16), indicating that “connectivity stiff, single-cell sloppy” is reproducible across independent implementations of the same organism rather than an artefact of one model (Supplementary §6.6).

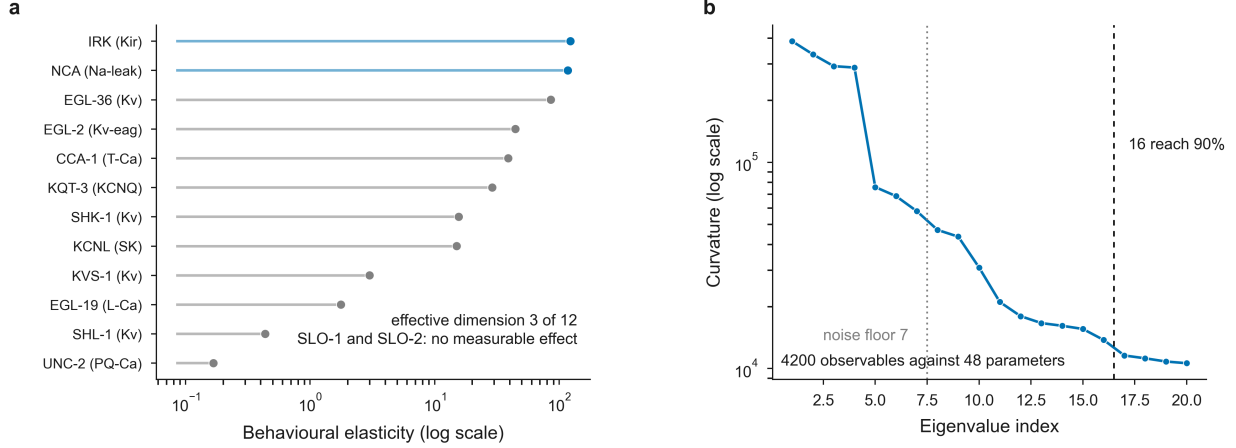


Figure 4: Stiff and sloppy structure at named-channel and higher-dimensional scale. a Behavioural elasticity of the BAAIWorm named conductances, read through the full motor-neuron trajectory, ordered from stiff (top) to sloppy (bottom). The stiffest are the inward rectifier IRK (123.5), the sodium leak NCA (117.7) and the potassium conductances EGL-36 (86.0) and EGL-2 (44.6); the sloppiest are SHL-1 (0.44) and the fast voltage-gated calcium channel UNC-2 (0.17), with EGL-19 (1.77) close behind. The calcium-activated potassium channels SLO-1 and SLO-2 produce no measurable change and are omitted. Effective dimension 3 (90%) / 6 (99%) over the twelve mechanisms with a measurable effect. **b** Eigenspectrum of the behavioural Gauss–Newton curvature of the 48 FlyGym actuator gains, measured through 4200 joint-angle observables rather than ten summary scalars. Effective dimension 16 (90%) / 38 (99%) of 48, against a noise floor of 7 / 22 obtained from seed-to-seed variation at identical parameters. Behaviour is largely insensitive to the fast voltage-gated calcium conductances and blind to the calcium-activated potassium conductances, which are the currents known to be homeostatically compensated and variable between animals. The FlyGym figure moves from 1 to 16 when the observable set is enlarged, so an effective dimension measured through a small observable set is a statement about the measurement. Curvature is the Gauss–Newton form $H = J^T W J$, computed by central finite differences and therefore positive semi-definite.

Convergent observations in independent biological data

Four independent datasets, none of which involves our forward models, exhibit low-dimensional or graded stiff/sloppy structure of the same kind (Fig. 5). They compare different levels of description—model-parameter curvature, neuronal phenotype variability, protein-sequence conservation and cross-individual behavioural variation—and agree at the level of dimensionality and direction. (1) *Pyloric degeneracy*. Applying our Hessian probe to the Prinz–Marder population of conductance-based models that produce the same triphasic pyloric rhythm (1) (rhythm-validity gated; valid rate 0.56%) yielded a 31-parameter eigenspectrum spanning approximately ten orders of magnitude with effective dimension 2.13/31: only 2–6 stiff directions, with stiffness on the leak and inhibitory-synapse conductances and sloppiness on the fast sodium conductances, the currents known to be compensated in this circuit. This canonical degeneracy dataset exhibits the same low-dimensional stiff/sloppy partition as the simulators. (2) *Neuronal property variability*. Across $n=1914$ Allen cortical neurons, electrophysiological properties span a stiff/sloppy gradient (coefficient of variation 0.08–1.85: resting potential stiff, adaptation and firing rate sloppy) on a property-space manifold of participation dimension 4.78/11. (3) *Gene conservation*. Across 40

C. elegans ion-channel and synaptic genes, ortholog identity across *Caenorhabditis* ranges from 77.9 to 99.9%; the most conserved is the connectivity gene *unc-9* (gap junction, 99.9%), consistent with the connectivity axes being stiff in both worm models. (4) *Behavioural variation*. Across $n=61$ isogenic N2 worms (683 Tierpsy features), cross-individual behaviour occupies a participation dimension of 9.75; genetically identical animals vary on an approximately ten-dimensional behavioural manifold, the empirical counterpart of the parameter manifold. The directions that biology conserves and those it permits to vary are ordered consistently with the models' identifiable and degenerate axes, at the level of low-dimensionality and qualitative direction.

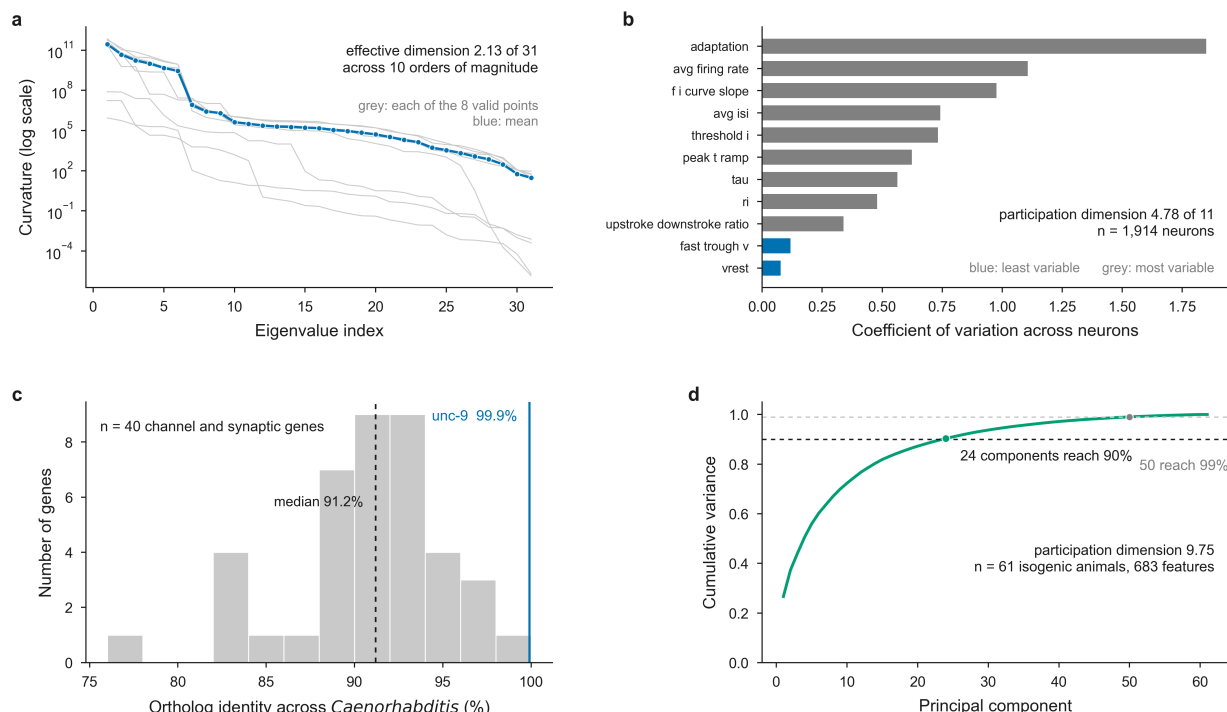


Figure 5: Low-dimensional stiff and sloppy structure in four biological datasets that use none of the simulators. **a** Prinz–Marder pyloric model population, re-gated to the canonical triphasic rhythm and probed with the identical Hessian-eigenspectrum method. Effective dimension 2.13 of 31 conductances across a spectrum spanning approximately ten orders of magnitude; stiff, leak and inhibitory-synapse conductances; sloppy, fast sodium conductances. Gating: all three cells bursting, period in [500, 2000] ms, strict AB/PD<LP<PY phase ordering; valid rate 0.56%; $K=8$ valid points, confirmed by an expanded sweep of 151 valid points at 2.4 ± 0.9 (Supplementary §2.1). **b** Electrophysiological property variability across Allen Cell Types cortical neurons. Coefficient of variation spans 0.08 (resting potential, stiff) to 1.85 (adaptation, sloppy); the property-space manifold has participation dimension 4.78 of 11; $n=1914$ neurons. **c** Ortholog identity of 40 *C. elegans* ion-channel and synaptic genes across *Caenorhabditis*, ranging from 77.9 to 99.9%. The gap-junction gene *unc-9* is the single most conserved at 99.9%. **d** Cross-individual behavioural manifold of isogenic N2 worms, 683 Tierpsy features, $n=61$ animals; participation dimension 9.75, with 24 principal components reaching 90% of the variance and 50 reaching 99%. All four datasets are public and no simulator is used in any of them; the comparison is made at the level of low dimensionality and direction, not of individual eigenvectors.

Discussion

We report a measurement of what behaviour can and cannot identify in whole-organism nervous-system models. In the nematode model built on the connectome, behaviour constrains eight of three thousand connection directions, and the conductances it cannot see are the fast voltage-gated calcium and calcium-activated potassium currents. Measuring further aspects of one action adds little. In both systems the behaviours compared are aspects of a single action—descending turn commands driving one pattern generator in the fly, mechanosensory touch at two sites and two intensities in the nematode—and the subspaces they determine overlap far more than they complement, with the gain saturating after three or four. The overlap is itself the finding: aspects of an action are largely redundant for identifying parameters, and their overlap measures degeneracy without choosing a threshold. Whether genuinely different actions (chemotaxis, escape, feeding) behave the same way is untested: no simulator available to us implements more than one in a form that can be driven independently. That is a requirement on the models before it is a question about the animals.

The organising principle is that behaviour reports on a small, structured part of a model. Any one behaviour pins a few stiff parameter combinations, and different behaviours pin the combinations their function requires. Between aspects of one action the subspaces overlap heavily and the union saturates at once; a shuffled control in which each behaviour’s Jacobian is scrambled gives a flat and higher value, so the measured low dimensionality is structure and not an artefact of the estimator.

This recasts degeneracy as a measurable resource rather than a fixed limit. A connectome-constrained simulator is identifiable up to the diversity of behaviours elicited, which makes assay diversity, rather than recording duration, the quantity to optimise. The directions that remain after the whole repertoire has been used, the residual degenerate core, are the candidate neutral directions.

The result lifts the sloppy-models picture (3, 4, 35) and pyloric degeneracy (1, 6) from single circuits to behaving whole organisms. It also explains why behaviour-based calibration recovers a working parameter set without recovering unique biophysical values: it recovers the stiff union, not the core. The independent datasets indicate that the unconstrained directions tend also to be the more variable and less conserved ones in biology, consistent with the biophysical diversity documented within identified cell types (44).

A second connectome-constrained model, in a different phylum, gives the same picture. The flyvis optic-lobe model of *Drosophila* has 734 free parameters in three groups: biases, membrane time constants and synaptic strengths. Probed across all three, its behavioural curvature has effective dimension 14 (90%) / 32 (99%) of 330 sampled parameters, and its response is proportional to the step across two decades (Supplementary §5.7). A visual and a motor circuit thus agree that behaviour reaches a few per cent of a connectome-scale parameter space.

How much was recorded sets what can be found, and by how much is now measurable. A Gauss–Newton curvature built from n_{obs} observables has rank at most n_{obs} , and the models track that bound: reading the visual model through fewer output channels moves its effective dimension almost one for one with the channel count, and the FlyGym figure moves from 1 to 16 when six summary scalars are replaced by 4200 joint angles. An effective dimension reported through fewer observables than parameters is therefore a property of the assay. We report effective dimensions only where observables outnumber parameters, and give the observable count, the step-size scan and the noise floor with every figure; where those conditions hold, the nematode and the fly agree.

Three limitations bound these results. The choice of behavioural observables is not unique, and a poorly chosen set could understate the identifiable rank. We therefore re-measured with 7,200–

14,400 posture observables in place of six summary kinematics and recovered the same effective dimension of 2–3 of 7 (Supplementary §6.2), and we verified the estimator itself against constructed spectra of known rank (Supplementary §6.8). The per-channel decomposition reaches named single conductances but still scales each conductance type uniformly across cells rather than cell by cell, and the mapping from parameter axes to biophysical classes is biophysically informed rather than learned. Two independently constructed worm models nevertheless agree on the class-level partition, which indicates that the result is not an artefact of one parameterisation. The alignment between model stiff axes and the biological datasets is established at the level of low dimensionality and qualitative direction rather than per-eigenvector correspondence, and the behaviour-to-neural prediction that would close the loop is confirmed only in gross statistics and across modalities (Supplementary §5.4). Behaviour also does not uniquely localise which mechanism was perturbed, since the detectable mechanisms are mutually confusable and the fast calcium channels are behaviourally near-silent (Supplementary §5.5).

The strongest outstanding test is prospective: predicting from the model which perturbations should produce the largest behavioural change, and testing that ordering on data not used to build it. Such a test requires perturbations resolved at the coupled parameter combinations the geometry identifies, not at single genes or channels, and such datasets do not yet exist. Defining those endpoints in advance also protects the statistics, which matters when independent samples are few.

Together, these results show that behavioural identifiability is a quantity that can be measured rather than assumed, and that its extent is set by the diversity of behaviours elicited. For digital-organism programmes this gives a division of labour: the stiff subspace is the part of a fitted model that behavioural data pin down and that can be read as biology, and the free directions name the parameters for which physiological measurement remains essential. The same geometry prices an experiment before it is run—an assay that adds another aspect of the same action re-measures known combinations, while one that changes what the circuit must do reaches new ones. The framework applies to any mechanistic simulator fitted to behaviour, and it turns the question "what does this model's fit actually pin down?" into one that can be answered before the fitting is done.

Methods

Manifold drift. For each simulator, N optimisations (separable CMA-ES or SPSA) from independent starts are fitted to one fixed behavioural target; we report behavioural recovery, the normalised per-dimension dispersion of the N recovered parameter vectors, and the PCA participation dimension of the solution cloud. **Behavioural curvature.** All curvature reported here is the Gauss–Newton form $H = J^\top W J$, built from the finite-difference observable Jacobian $J_{ik} = \partial \text{obs}_i / \partial \theta_k$ and a diagonal positive semi-definite whitening matrix W of inverse per-observable scales. H is therefore positive semi-definite by construction and equals the Fisher information of the parameters given the behavioural observables, so its eigenvalues are non-negative and its effective dimension is the local practical-identifiability rank in the standard sense (3, 33). Eigenvalues are obtained by direct eigendecomposition (low dimension) or Lanczos iteration (high dimension). The effective dimension at level q is the smallest k such that $\sum_{i \leq k} \lambda_i \geq q \sum_i \lambda_i$ with eigenvalues sorted in decreasing order; we report $q = 0.90$ and $q = 0.99$, together with the threshold-free participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ (Supplementary §6.9). Curvature is evaluated at the published operating point, and its point dependence is measured directly: on the connectome-scale model, four displaced operating points give effective dimension 8–12 with stiff subspaces coinciding across points

at cosine 0.89 against a random null of 0.15 (Supplementary §5.3); on modWorm, five displaced optima and twenty-four random points all give a low effective dimension (Supplementary §6.8). A direction counts as flat when λ falls to the finite-difference floor; residual eigenvalues of magnitude below that floor, of either sign, are numerical noise and are treated as zero. We adopt it as the primary cross-simulator metric because it is robust to the choice of observable and parameterisation, unlike a raw perturbation-sensitivity ratio, and because the manifold-drift probe provides an independent, optimisation-based cross-check that agrees with it ($\rho=0.96$). **Eigenworm identity.** Real *C. elegans* skeletons (OWMD N2) and model body postures are reduced to tangent-angle eigenmodes by PCA; we report the posture effective dimension, the top-four cumulative variance, and the principal-angle cosine between model and real eigenworm subspaces, with significance assessed against random subspaces of matched dimension. **Complementary tiling and saturation.** Per-behaviour stiff subspaces are taken as the leading Hessian eigenvectors under each behavioural target; we report pairwise stiff-subspace cosines, the union effective dimension over a behaviour set, and the saturation curve of union dimension against the number of behaviours. Context dependence (BAAIWorm) is the ratio of a parameter class’s behavioural sensitivity between two behaviours (chemotaxis versus weak-gradient locomotion). The same union/saturation construction is applied across species (fly larva, larvaworld; rodent, Virtual Rodent / dm_control). **Cell-scale ground truth.** Eleven published cell models (five neuronal, six biochemical oscillators) are measured with the identical estimator and recovered from random starts $\theta^* e^{\mathcal{U}(-\ln f, \ln f)}$ at $f=1.5-10$, with a phase-invariant feature loss computed at full resolution; a fit counts as converged below behavioural distance 0.02 and only converged fits contribute parameter errors (Supplementary §7). **Calibration.** Per-dimension rank-based simulation-based calibration (45, 46) of an amortised neural posterior; a dimension passes when the per-axis Kolmogorov–Smirnov test is non-significant at $\alpha=0.05$. **External datasets.** The Prinz–Marder pyloric model population re-gated to the triphasic rhythm and probed with the identical Hessian-eigenspectrum method; Allen Cell Types electrophysiology; WormBase ParaSite / Ensembl Metazoa ortholog identities; and OpenWorm Movement Database isogenic-N2 Tierpsy features. All external data are public, and no simulator is used in datasets (1)–(4). Full specifications, cohort sizes and reproduction commands are given in the Supplementary Information.

Data availability

All datasets analysed in this study are publicly available. The external biological references are: the Prinz–Marder conductance-based model population of the stomatogastric ganglion (1); the Allen Cell Types electrophysiology database (<https://celltypes.brain-map.org>); ortholog identities from WormBase ParaSite and Ensembl Metazoa (<https://parasite.wormbase.org>); and the OpenWorm Movement Database (<http://movement.openworm.org>). The whole-organism simulators are the published releases of BAAIWorm (8), modWorm (9), flybody (10), NeuroMechFly v2 / FlyGym (11), the connectome-constrained optic-lobe model of Lappalainen et al. (47), larvaworld, and the Virtual Rodent of the dm_control suite. All quantities reported in the main text and Supplementary Information are provided as result files, one file per analysis as named in each Supplementary section, in the deposit at <https://github.com/ChenxiHeCam/behavioural-parameter-manifold>, archived at Zenodo (<https://doi.org/10.5281/zenodo.21594740>).

Code availability

All analysis code, the per-experiment result files, and a container that reproduces every reported quantity and figure from those files are available at <https://github.com/ChenxiHeCam/behavioural-parameter-> under an MIT licence, archived at Zenodo (<https://doi.org/10.5281/zenodo.21594740>). The reproduction entry point verifies each numerical claim against its source file and regenerates the figures.

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Author contributions

C.H. conceived the study, designed and performed all analyses, and wrote the manuscript.

Competing interests

The author declares no competing interests.

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